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RESEARCH

Human impacts on fish biodiversity in the Yangtze River Basin: a multidimensional assessment using the Cumulative Change in Biodiversity Facets (CCBF) index

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Abstract Human activities are causing a severe decline in fish species diversity globally. Although biodiversity is inherently multidimensional, most previous studies have concentrated on taxonomic changes, neglecting functional and phylogenetic dimensions, therefore limiting our understanding of biodiversity changes. The Yangtze River, an ecologically and economically critical system, supports high fish biodiversity and important fisheries. Yet, the spatio temporal patterns and anthropogenic drivers of biodiversity changes over the past decades remain unclear. We compiled a comprehensive fish occurrence dataset covering 56 geographic units across the Yangtze River Basin for historical (1976–2010) and current (2011–2020) periods. We assessed taxonomic, functional, and phylogenetic richness and dissimilarity, and quantified overall changes using the Cumulative Change in Biodiversity Facets (CCBF) index. Over 80% of units declined in all three richness facets, while dissimilarity generally increased, particularly in the middle and lower mainstem. CCBF values (0–12) that reflect the overall magnitude of biodiversity reorganization were highest in the middle reaches, with the Mainstream region experiencing the strongest biodiversity changes between historical and current periods. Changes were primarily driven by human disturbances, including river fragmentation, consumptive water use, and urbanization. These declines in multidimensional biodiversity may reduce ecosystem resilience and fishery productivity, and

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threaten the long-term sustainability of fish stocks. The loss of functional and phylogenetic diversity could especially weaken the capacity of fish communities to withstand environmental fluctuations and recover from disturbances. Our findings highlight the urgent need to integrate taxonomic, functional, and phylogenetic dimensions into conservation strategies—such as trait-based restoration and phylogenetically informed protected areas—to enhance resilience and mitigate biodiversity loss in large river systems.

Keywords Fish · Taxonomic diversity · Functional diversity · Phylogenetic diversity · Human activities · Yangtze River Basin

Introduction

Human activities have triggered a severe and widespread decline in global biodiversity, with freshwater ecosystems being among the most threatened. The Yangtze River Basin, the third biggest river basin globally, is a critical global biodiversity hotspot (Chen et al. 2020). With a total drainage area of 1.8 million km^2 and a mainstem length of nearly 6300 km, the basin covers 18.8% of China's land area and is home to almost 440 million people, contributing 40% of the national GDP (Zhang et al. 2020a). It supports over 400 freshwater fish species, including numerous endemics (Yang et al. 2023) and contains more than 4000 lakes, providing irreplaceable ecological and economic services (Cao 2017; Wang et al. 2018). However, rapid economic development has subjected the basin to an array of anthropogenic pressures—including dam construction, river fragmentation, overfishing, water pollution, and urbanization placing its aquatic ecosystems under unprecedented threat and raising concerns about potential ecological collapse (Jia et al. 2020; Xiong et al. 2021). In response to this critical situation, the Chinese government implemented a ten-year full fishing ban on the Yangtze River in 2021, as an attempt to restore ecosystem integrity (Mei et al. 2020). Quantifying the efficacy of this bold conservation action requires a robust, multidimensional baseline of fish biodiversity state immediately prior to the ban. Our study aims to establish this baseline by assessing biodiversity changes from 1976 to 2020, without directly testing the ban's effects (as the ban occurred after our study period). To capture the full ecological transition from pre-dam to post-dam conditions, we set the temporal boundary at 2010, when the Three Gorges Dam reached its designed full operational level of 175 m, following a phased impoundment that began in 2003 (Ministry of Water Resources & National Development and Reform Commission, 2020). This breakpoint also reflects the shift from less systematic historical surveys to more standardized contemporary monitoring, ensuring robust comparability between the two periods.

Traditionally, biodiversity benchmarks were established using taxonomic indicators such as the richness in species (Chen et al. 2009; Ye et al. 2014). For instance, Chen et al. (2009) showed taxonomic richness in the Yangtze River declined by over 30% between the 1980s and early 2000s, primarily due to overfishing and habitat degradation. Moreover, Ye et al. (2014) reported that species losses were particularly severe in the middle and lower reaches of the mainstream, where anthropogenic pressures were most intense. While fundamental, this taxonomic approach provides an incomplete picture, as it fails to capture the ecological roles species perform (functional diversity) or their distinct evolutionary histories (phylogenetic diversity) (Brosse et al. 2021; Schweiger et al. 2018). For instance, local species extirpations reduce taxonomic, functional, and phylogenetic richness, whereas non-native fish establishment can offset these declines or even increase functional or phylogenetic richness, as a slight change in the number of species may involve the loss or gain of unique traits or lineages (Xiong et al. 2023). This has been demonstrated for many river basins across the globe, where modest taxonomic changes led to drastic increases in functional diversity due to the introduction and subsequent establishment of non-native species with original traits compared to those of the native fauna (Toussaint et al. 2018). In addition, local extirpations and introductions can generate a wide array of regional responses ranging from homogenization to differentiation (Toussaint et al. 2014; Villéger et al. 2011), through the spatially heterogeneous loss of native species and introduction of non-natives. Such changes were in some cases drastic due to the loss of species with unique traits and their replacement by a few, functionally similar, non-native species, which led to drastic functional homogenization of historically distinct faunas (Keck et al. 2025; Su et al. 2021).

Urbanization, consumptive water use, and river fragmentation by dams are among the most pervasive anthropogenic pressures reshaping freshwater fish

assemblages globally, yet their mechanistic effects on different biodiversity dimensions are often context-dependent. Urbanization typically reduces all three facets of alpha diversity through habitat loss, water quality degradation, and flow regime alteration, while simultaneously promoting the invasion of non-native species that may homogenise formerly distinct assemblages (Petsch et al. 2021; Li et al. 2024). A recent study in the Yangtze River Basin found that hydrology and water quality—both strongly influenced by land use change—were the key drivers of individual biodiversity indices for fish, with fish showing the strongest responses to multiple stressors among all biological groups (Li et al. 2024). Consumptive water use for agricultural, industrial, and domestic purposes reduces river discharge and seasonal flows, filtering out species requiring high flow velocities or specific hydraulic habitats, thereby decreasing functional and phylogenetic richness while increasing functional dissimilarity among units where water extraction pressures differ (Döll et al. 2009; Su et al. 2021). River fragmentation by dams acts as physical barriers to fish migration, disrupts longitudinal connectivity, and alters sediment and nutrient transport, leading to the loss of migratory species and the isolation of upstream populations. Recent research has demonstrated that physical variables such as water temperature, specific conductance and total dissolved solids—all modified by dam operations—are key factors influencing both taxonomic and phylogenetic diversity of fish in the Three Gorges Dam area, with environmental filtering playing a dominant role in fish community assembly (Zhang et al. 2025; Zhu et al. 2025). Fragmentation can increase beta diversity through spatial differentiation among isolated fragments, or conversely, promote homogenization, where impoundment creates uniformly lentic conditions across previously heterogeneous reaches (Villéger et al. 2011; Adams et al. 2025). Based on these mechanisms, we made the following explicit predictions: (1) urbanization and consumptive water use will exhibit strong negative correlations with all three facets of alpha richness across the Yangtze River Basin; (2) the degree of fragmentation will be positively associated with beta dissimilarity, reflecting increased spatial heterogeneity due to barrier effects and the loss of migratory species; and (3) the Mainstream region where dam density and urbanization are highest, will exhibit the largest cumulative biodiversity change (highest CCBF scores), with the Lake region experiencing comparatively lower magnitudes of change due to stronger hydrological connectivity and pre-existing wetland reserves.

Therefore, an integrative framework that simultaneously quantifies changes in taxonomic, functional, and phylogenetic facets of biodiversity is required to understand the anthropogenic impacts on the Yangtze's fish assemblages. To address this critical gap, we applied the Cumulative Change in Biodiversity Facets (CCBF) framework (Su et al. 2021) to quantify temporal changes across six biodiversity facets—taxonomic, functional, and phylogenetic richness and dissimilarity—between the historical (1976–2010) and current (2011–2020) periods. Unlike temporal β-diversity, which captures only compositional turnover between two periods, the CCBF index integrates both alpha and beta changes across taxonomic, functional, and phylogenetic facets. This multidimensional approach detects hidden changes (e.g., functional homogenization despite stable taxonomic richness) and provides a standardized composite score (0–12) reflecting the overall magnitude of biodiversity reorganization. Briefly, this framework provides a score ranging from zero, if no change occurred in any of the 6 facets, to 12 if strong changes occurred in all six facets. This framework, initially developed on a global scale was here adapted to the 56 geographic units of the whole Yangtze River Basin. Here, we show the spatial pattern of CCBF across 56 geographic units to provide a comprehensive assessment of how anthropogenic pressures have reshaped fish assemblages immediately prior to the fishing ban. We then disentangled the contribution of the six biodiversity components of the CCBF to biodiversity changes, as distinct processes can support a given CCBF score. Finally, we investigated the extent to which fisheries, non-native species introduction, consumptive water use, urbanization, dam construction, and water pollution contributed to the biodiversity changes.

Methods

Study area

The Yangtze River is located on the eastern coast of China (120° E–122°30′ E, 32°30′ N–29°30′ N). For spatial analysis, the Yangtze River Basin was divided into 56 geographic units following (Kang et al. 2018), with each unit having an annual

discharge area exceeding 3.0×10^4 km^2 (Fig. 1). These units were grouped into three region types—Mainstream, Tributary, and Lake, which are based on water discharge, river length, and habitat characteristics (Fig. 1; see Fig. S1 for the delineation of 11 sub-basins; Changjiang Water Source Committee, 1999). Within this framework, the Three Gorges Dam (surface area 1,084 km^2)—one of the largest hydropower projects globally—is located in the Middle Mainstream region near Yichang (approximately 111.0° E, 30.8° N), representing a major hydrological threshold that has profoundly influenced fish habitats in the basin (Zhang et al. 2020b).

Data collection

Occurrence data

The freshwater fish occurrence dataset for the Yangtze River Basin was compiled from multiple sources, including a systematically reviewed dataset from a prior basin-wide study (Kang et al. 2018), our field surveys, and an extensive review of the literature (Table S1). To evaluate the impact of major anthropogenic pressures on biodiversity, we defined two distinct temporal periods separated by a critical hydrological milestone: a historical baseline (1976–2010) preceding the completion of the Three Gorges Dam to its full operational level, and a current

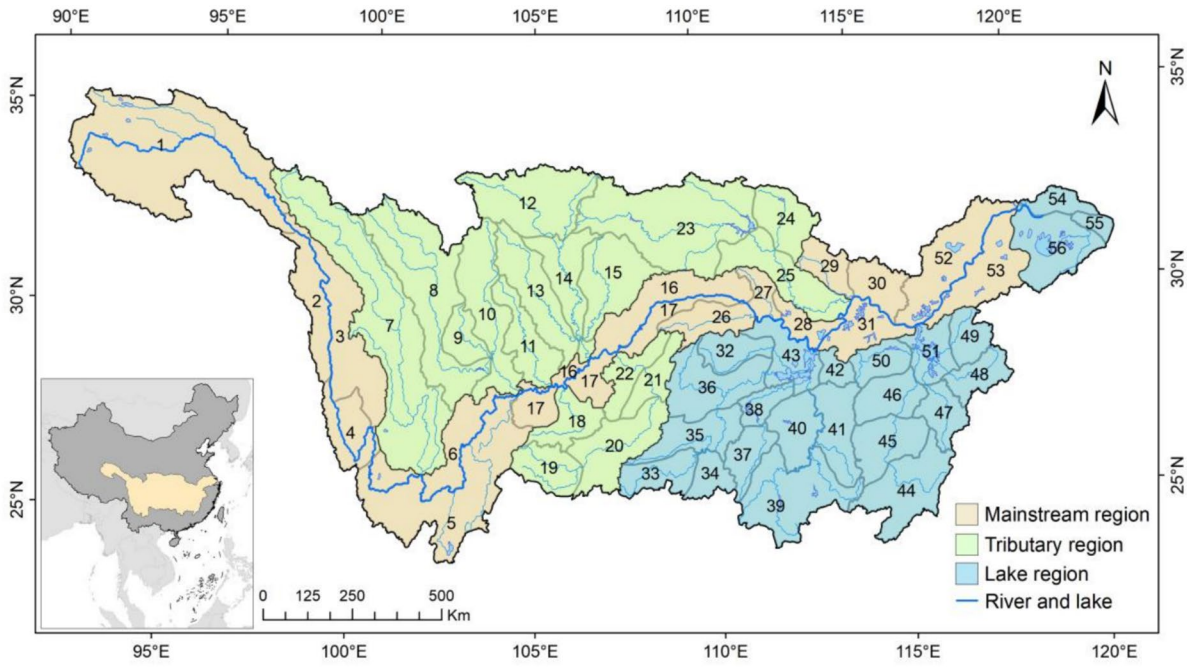


Fig. 1 Location of Yangtze River Basin, geographic regions, and units. "The 56 geographic units: 1, Tongtian River; 2, Eastern Tibet; 3, Baiyu-Batang; 4, Suoduogang River; 5, lower Jingsha River (right bank); 6, lower Jingsha River (left bank); 7, Yalong River; 8, Dadu River; 9, Qingyi River; 10, Min River; 11, Tuo River; 12, upper Jialing River; 13, Fu River; 14, middle-lower Jialing River; 15, Qu River; 16, upper Yangtze (left bank); 17, upper Yangtze (right bank); 18, Chishui River; 19, upper Wu River; 20, middle Wu River; 21, lower Wu River (before the Pengshui hydropower station); 22, lower Wu River (behind the Pengshui hydropower station); 23, upper Han River; 24, Tang and Bai Rivers; 25, middle-lower Han River; 26, Qing River; 27, Ju and Zhang Rivers; 28, four lakes in Hubei Province; 29, Yun and Huan Rivers; 30, streams in the northeast Hubei Province; 31, streams in west Hubei Province; 32, Li River; 33, upper Yuan River (Qingshui River); 34, upper Yuan River (Qu and Wu Rivers); 35, middle Yuan River; 36, lower Yuan River; 37, upper Zi River; 38, middle-lower Zi River; 39, upper Xiang River; 40, middle-lower Xiang River (left bank); 41, middle-lower Xiang River (right bank); 42, Dongting Lake in Northeast Hunan Province; 43, Dongting Lake plain; 44, upper Gan River; 45, middle Gan River; 46, lower Gan River; 47, Fu River; 48, Xin River; 49, Rao River; 50, Xiu River; 51, Poyang Lake; 52, Chao Lake; 53, Shuiyang and Qingyi Rivers; 54, streams in Jiangsu Province; 55, Chongming Island; 56, Tai Lake and the delta'' from Kang et al. (2018). The yellow, green, and blue regions represent the Mainstream, Tributary, and Lake regions, respectively

period (2011–2020) after the reservoir reached stable post-impoundment conditions. This division is ecologically relevant because the attainment of 175 m level around 2010 marked a significant and lasting shift in the river's flow regime and physical habitat structure, enabling a clear assessment of its cascading effects on fish assemblages. We explicitly account for the dual processes of native species disappearance and non-native species introduction to understand their compound effects on multiple biodiversity facets. To mitigate potential biases from differing sampling efforts between periods, we prioritized data from standardized ecological surveys and consistently applied the same geographic units across both periods, ensuring robust comparability.

Species location records were identified at geographic units to create the species distributional data matrix (species × units) in which species presence or absence in each geographic unit was represented by "1" or "0", respectively. All taxonomic names were verified using FishBase (Froese and Pauly 2018) to remove synonyms and invalid species. A total of 369 and 259 species were collected for the historical and current periods, respectively (Supplemental data). Although the dataset contains introduced exotic species, the primary aim of this study was to evaluate changes in whole-assemblage diversity in response to anthropogenic pressures.

Functional traits

Recognizing that the morphological, behavioral, and physiological dimensions of ecological niches serve as effective proxies for ecological performance, a total of 14 functional traits matrix (species × traits) was built based on three categories: morphological parameters (measurements of head, eye, fin, body, etc.), habitat type (water column position, water temperature, and water zone), and trophic guild (trophic level and feeding habits) (Toussaint et al. 2016; Villéger et al. 2011). Nine morphological traits associated with fish locomotion and food acquisition, along with five habitat-related traits, were primarily gathered from the FISHMORPH database (Brosse et al. 2021). Additional data for these traits were obtained from FishBase (Froese and Pauly 2018), relevant literature, and Chinese fish books (Lin et al. 2023). Morphological traits indicate various ecological niche aspects, such as swimming behavior and habitat use (Goldstein and Meador 2011). Trophic traits are vital for understanding food webs and predator–prey dynamics (Elliott et al. 2007). Habitat traits define nursery, feeding, migratory, and refuge areas, with species distribution constrained by water body properties (Potter et al. 2013). Water zone accounts for the main environment occupied by the species (river, lake, marine, and stream environments), the water column position defines the primary living layer utilized by the species (benthic, pelagic, or surface zones), and the water temperature delineates the thermal range within which the species predominantly thrives (tropical, sub-tropical, or temperate zone) (Kang et al. 2018). The detailed information for these functional traits is shown in Fig. S2.

For numerical traits, we used the mean value when multiple records were available for a species (Siefert et al. 2013). Categorical traits were coded into discrete categories based on thermophily, water-column position, habitat guild, and trophic guild. We then computed a Gower distance matrix to quantify pairwise trait dissimilarities among species using both continuous and categorical variables. Functional space on the first five principal coordinate axes was then constructed using principal coordinate analysis (PCoA), creating a five-dimensional function space that explained more than 80% of the overall variation. Intraspecific variation in functional traits was not considered in this study, as its influence on functional diversity is negligible at the regional spatial scale analysis (Toussaint et al. 2018).

Phylogenetic diversity

FishPhyloMaker R package was used to construct a phylogenetic tree for Yangtze freshwater fish (Nakamura et al. 2021; Xu et al. 2024). The package merged and pruned species from a comprehensive backbone phylogenetic tree to create a phylogenetic tree for 367 species (Rabosky et al. 2018). A total of 2 species names were excluded as unidentifiable during the process. Consequently, we obtained a clean occurrence dataset with phylogenetic information for a final set of 367 species (343 native and 24 non-native) in the Yangtze River Basin. On the basis of this phylogenetic tree, the pairwise phylogenetic distances across all species were calculated using the cophenetic function of the *ape* R package.

Measuring temporal changes in fish biodiversity

Diversity indices were assessed for all 56 geographic units, with each unit having more than five fish species occurrence data. For the current and historical periods, the six facets of biodiversity for the fish assemblages from each of the 56 geographic units were evaluated: the number of species is the measure of taxonomic richness (TR), the volume of the functional space occupied by an assemblage is the measure of functional richness (FR) (Villéger et al. 2008), and the length of all the branches connecting every species in the assemblage represented by the phylogenetic tree is the measure of phylogenetic richness (PR) (Faith 1992; Su et al. 2021). The dissimilarity between assemblages (i.e., beta diversity) was considered in addition to these indices that describe variety within each collection (i.e., alpha diversity). Beta diversity, quantified as the dissimilarity index (DI) based on the Jaccard metric for taxonomic, Gower distance for functional, and UniFrac for phylogenetic facets (Su et al. 2021). The taxonomic, functional, and phylogenetic dissimilarity for each specific unit was then determined by averaging its pairwise dissimilarity values with all other units, their dissimilarity indices were calculated using the *betapart* R package.

Finally, the temporal change (δDI) was determined by taking the ratio $DI_{current}$ to $DI_{historical}$ for each of six diversity parameters in every geographical unit. Across all units, we then calculated the median of all δDI values less than 1, and the median of all δDI values greater than 1. A score was assigned as follows: if $\delta DI = 1$, the score is 0; if δDI is higher than the median of all the values lower than 1 and lower than the median of all the values higher than 1, it scores 1; and if δDI is lower than the median of all the values lower than 1 or higher than the median of all the values higher than 1, it scores 2. The six δDI scores for each unit were then summed to obtain a cumulative biodiversity change index (CCBF) ranging from 0 to 12 (Su et al. 2021). This cumulative index therefore considers taxonomic, functional and phylogenetic changes associated with species losses and gains in each geographic unit, and between each focal unit and all the others. Null values are virtual in the sense that they could only occur if all units maintained their historical taxonomic, functional, and phylogenetic compositions, since any alteration could affect all dissimilarity indices. We considered, according to Su et al. (2021), that CCBF scores ranging from 0 to 6 indicate moderate changes in biodiversity, either with all six facets scoring 0 or 1 or with up to three facets scoring 2. These values are considered indicative of moderate biodiversity changes. Strong changes in biodiversity are indicated by scores ranging from 7 to 12, which can affect all six facets or over 50% of the facets with a score of 2, as shown in Table S2 (Su et al. 2021).

Environmental and anthropogenic variables

Fourteen key environmental and anthropic variables (e.g., temperature, fishing intensity, hydrology) that potentially affect freshwater fish biodiversity were initially considered (Fig. S3). To control for multicollinearity among bioclimatic variables, six of the fourteen available were removed based on pairwise Pearson's correlations and variable connection descriptions ($|r| > 0.70$). The remaining eight variables were retained to characterize the primary ecological and anthropogenic drivers of native freshwater biodiversity: (1) native species richness (NSR); (2) river basin area (RBA), defined here as the area of each geographic unit; (3) the mean elevation of each geographic unit (DEM); (4) Strahler stream-order (ORD); (5) weighting the river order within each geographic unit by reach length (LEN); (6) urbanization index (URB); (7) degree of fragmentation (DOF); (8) consumptive water use (USE). Overall, these variables exhibited low pairwise correlations (|Pearson's $r| < 0.65$), indicating relative independence (Fig. S3).

NSR reflects the concept of the biotic resistance hypothesis, suggesting that species-rich assemblages exhibit greater resistance to disturbances (Levine 2000). NSR could also be considered a response variable; here we include it as a covariate to test for associations between baseline diversity and subsequent changes, following the analytical framework of Su et al. (2021). RBA data originated from a published global dataset (Pekel et al. 2016). DEM data were acquired from the Geospatial Data Cloud (https://www.gscloud.cn), and mean elevation per sub-basin was computed using the Zonal Statistics tool in ArcGIS 10.7. ORD was generated and hierarchically classified in the watershed river network using the Strahler method, implemented through the Hydrological Analysis tool of the software (Tarboton et al.

1991). LEN was subsequently calculated by weighting the Strahler stream order by reach length (Yang et al. 2024).

DOF, URB, and USE are well-known anthropogenic factors that affect the fish biodiversity in the Yangtze River Basin (Currie et al. 2004; Leprieur et al. 2011). DOF, indicating river network fragmentation, was sourced from a global 500 m resolution dataset (Grill et al. 2019). URB was developed from global nightlight data (DMSP-OLS version 4; Doll 2008). USE, representing water consumption, was calculated at 1 km resolution based on the formula $100 \times (\text{dnat} - \text{dant})/\text{dnat}$, using natural (dnat) and actual (dant) discharge data from the WaterGAP model (Döll and Siebert 2002; Grill et al. 2019). Finally, the grid data for DOF, URB, and USE were overlaid with our geographic units, and the average value within each unit was extracted for analysis.

Statistical analysis

The spatial simultaneous autoregressive error (SAR_{error}) model was used to quantify the relative contributions of NSR, RBA, DEM, ORD, LEN, URB, DOF, and USE to the CCBF values in three regions of the Yangtze River Basin (i.e., Mainstream region, Tributary region, and Lake region; Table S3), where all variables were available (Kissling and Carl 2008). These eight variables were first standardized to a zero mean and unit variance to guarantee equal weighting in the models. Nagelkerke's R^2 (Nagelkerke 1991) was employed as the pseudo R-squared to evaluate the performance of the models. Following model fitting, the Moran's *I* statistic was computed to check for broad spatial autocorrelation in the model residuals (Cliff and Ord 1981). This evaluation allowed for assessing the relative contributions of human activity and environmental processes to observed biodiversity changes. Prior to analysis, missing values in the trait database (constituting approximately 3% of the dataset) were imputed using the *Rphylopars* package by applying a phylogenetic generalized least squares model, which leverages phylogenetic covariance to estimate missing trait values. The R software environment version 4.2 was used for all statistical analyses (R Core team).

Results

Biodiversity changes in the Yangtze River Basin

Over the past decades, taxonomic richness (TR), functional richness (FR), and phylogenetic richness (PR) declined in more than 78% of the Yangtze River Basin units (Fig. 2A–C). The greatest declines in TR were observed in the Tongtian River (unit 1), Min-Tuo River (units 9–11), most of the Wu River (units 17–21), the middle reach of the Yangtze River (units 24–31-except 26), Gan River (46), Fu River (47), Xin River (48), and Poyang Lake (51), with a value below 0.5. FR and PR showed similar trends (Fig. 2A–C), but some discrepancies were also recorded. For instance, the Fu River (13) experienced strong TR and PR changes, but little variation in FR. Conversely, an increase was found in taxonomic dissimilarity (TD), functional dissimilarity (FD), and phylogenetic dissimilarity (PD) in more than 85% of the units (Fig. 2E–G). Changes in PR were slightly higher than those in TR and FR, whereas changes in FD were greater than those in TD and PD (Fig. 3). FD and PD showed substantial positive correlations with changes in TD, with Pearson's correlation coefficients of 0.79 and 0.95, respectively. The local biodiversity declines were significantly correlated to increases in dissimilarity for all facets (Fig. 3C). The inclusion or exclusion of non-native species did not significantly alter the overall patterns of change across the six facets of biodiversity in the Yangtze River Basin (Figs. 2, S3 and S4).

Cumulative change in biodiversity

The CCBF scores of all 56 units in the Yangtze River Basin were greater than or equal to 6 (Fig. 4A). The highest CCBF values were observed in the middle reach of the Jinsha River (units 3 and 4), the Tuo River (unit 11), the middle Wu River (units 20 and 21), the Han River (units 24 and 25), the middle reaches of the Yangtze River (units 26–31), the Xin River (48), and PoYang Lake (51), all reaching 12; they were primarily distributed in the Tributary region, accounting for 11%. The units with the lowest CCBF score of 6 in the Yangtze River Basin were mainly concentrated in the Tributary region, accounting for 16%. Relatively, the CCBF of the lower reaches of the Yangtze River (52–53 units) was

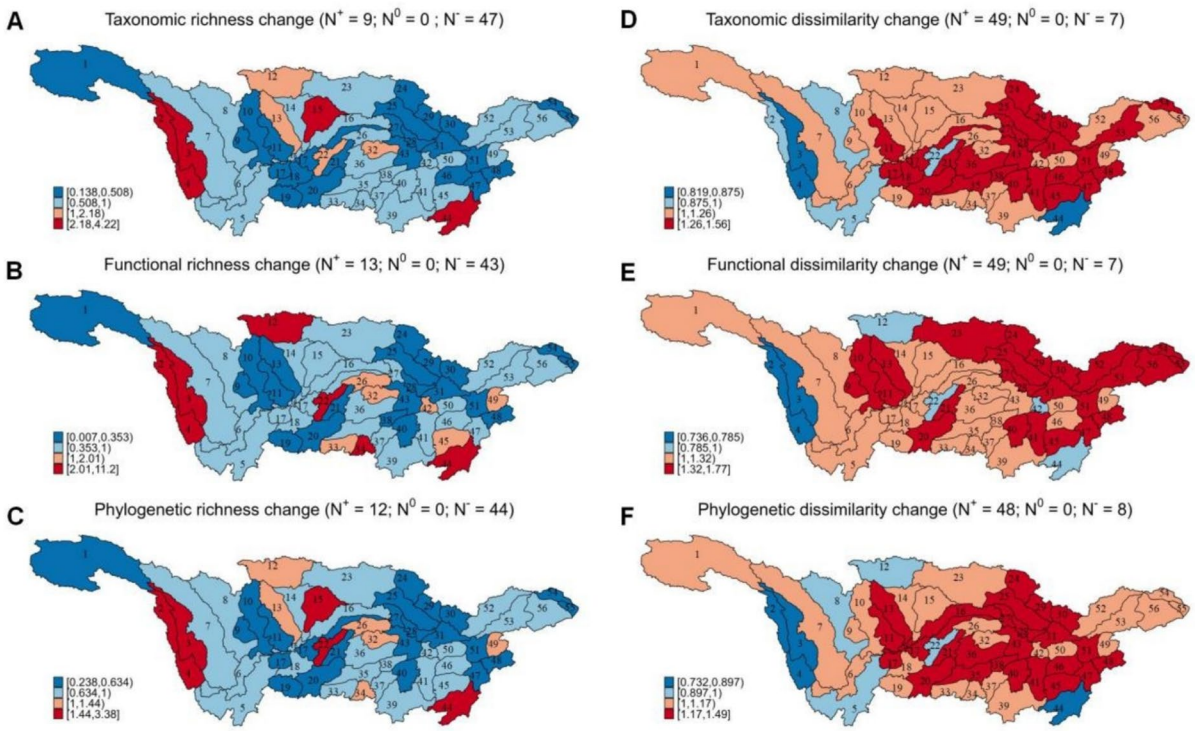


Fig. 2 Changes in each of the six biodiversity indices for Yangtze River Basin fish assemblages. Changes in taxonomic richness, functional richness, phylogenetic richness, taxonomic dissimilarity, functional dissimilarity, and phylogenetic dissimilarity are listed in order of appearance (**A**, **B**, **C**, **D**, **E**, and **F**). The original ratio of $DI_{current}/DI_{historical}$ is shown in legend values. At the top of each panel is the number of units where fish biodiversity increased (N^+), maintained unchanged (N^0), or decreased (N^-)

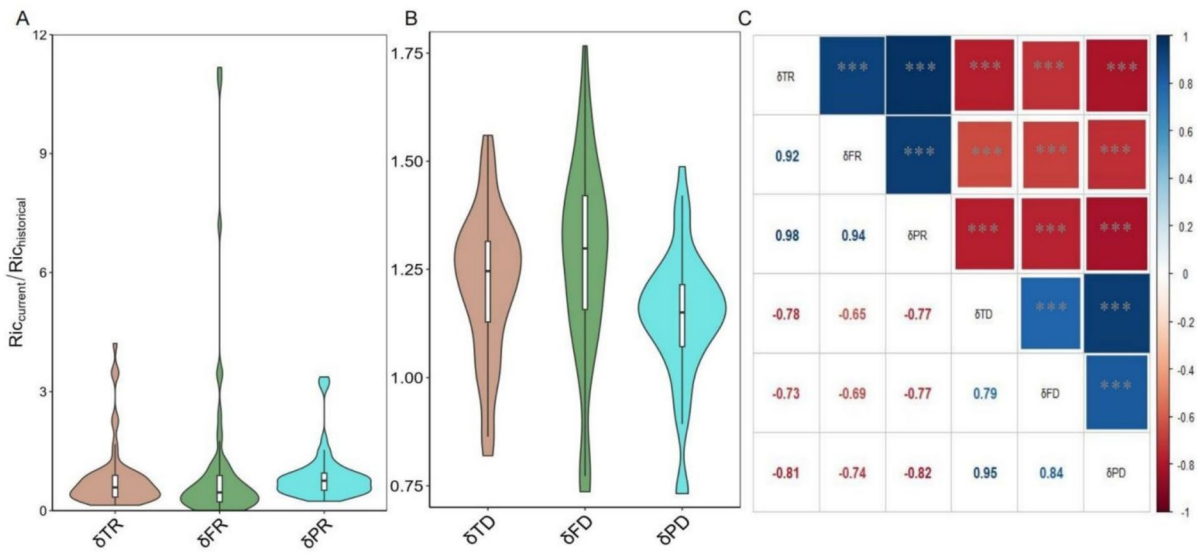


Fig. 3 Changes in biodiversity between current and historical periods. **A** and **B** Violin plots show the distribution of the three richness changes and three dissimilarity changes on the index's values, respectively. **C** Pearson's correlation between the changes in diversity indices (***$P < 0.001$)

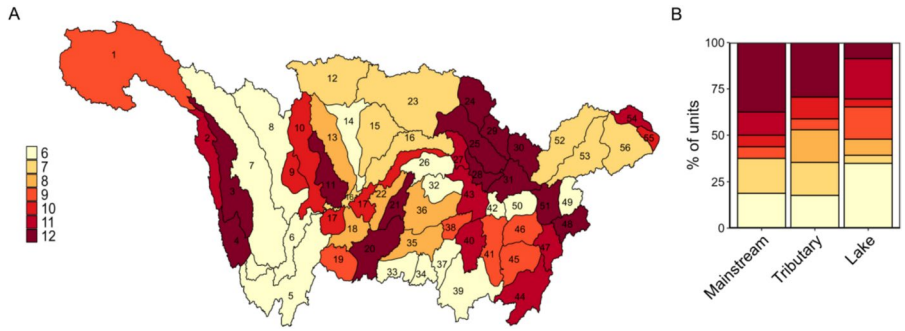


Fig. 4 Cumulative change in biodiversity of freshwater fish in 56 geographic units. A combined index considers the total changes in the six diversity indices, encompassing three dimensions (taxonomic, functional, and phylogenetic) determined at geographic unit scales. **A** Map showing changes in 56 units. **B** Percentage of units for seven intensities of change in each geographic region

smaller than that of the middle reaches (26–31 units) of the Yangtze River Basin. Fish assemblages from the Mainstream region had the largest changes in biodiversity, with more than 50% of the units achieving a CCBF score higher than 10 (Fig. 4B). Conversely, the Lake region experienced the smallest biodiversity changes, with more than 50% of the units reaching a CCBF score below 10. Overall, the biodiversity changes were highest in the Mainstream region (CCBF = 9.5 ± 2.6, mean ± SE), followed by the Tributary region (CCBF = 8.9 ± 2.4, mean ± SE), and then the Lake region (CCBF = 8.5 ± 2.3, mean ± SE, Fig. 4B).

Drivers of biodiversity change across the Yangtze River Basin

The Cumulative Change in Biodiversity Facets (CCBF) exhibited distinct, region-specific associations with anthropogenic and environmental drivers across the Yangtze River Basin (Fig. 5). At the basin scale, CCBF was positively correlated with human pressures, including urbanization (URB), the degree of fragmentation (DOF) and consumptive water use (USE). A clear divergence in primary drivers was observed among regions. In the Lake region, native species richness (NSR) and USE emerged as the principal factors positively associated with CCBF. In the Mainstream region, CCBF was negatively associated with NSR, while being predominantly driven by the degree of fragmentation (DOF). In the Tributary region, CCBF showed significant positive associations with DOF and URB, whereas length-weighted river order (LEN) showed a significant negative correlation. The relationship between NSR and CCBF displayed a contrasting spatial pattern, being positive in the Lake and Tributary regions but negative in the Mainstream region. Furthermore, other environmental factors, including river basin area (RBA), mean elevation of each sub-basin (DEM) and stream order (ORD), demonstrated no significant relationships with CCBF across all three regions.

Discussion

Fish biodiversity changes in Yangtze River Basin

A comprehensive understanding of the different components of biodiversity and the drivers of their change is essential for the effective conservation and restoration of biodiversity (Lazzari et al. 2020; Nicol et al. 2017; Yang et al. 2024). The present study not only assessed the degree of change in six key facets of fish diversity across the Yangtze River Basin but also used the CCBF index to evaluate their cumulative changes. The results show that all 56 sub-basin units within the Yangtze River Basin experienced substantial changes (CCBF > 6), revealing the profound and spatially distributed impact of environmental factors and human activities across the basin (Fig. 4A). Specifically, the most affected region is located in the middle reach of the Yangtze River, primarily due to a high degree of habitat fragmentation. This finding is consistent with previous research, indicating that the greatest impact on

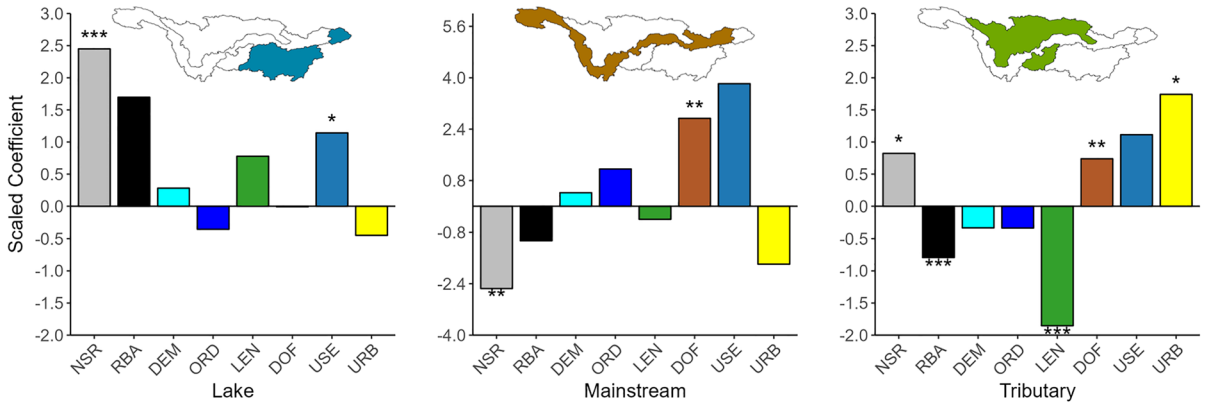


Fig. 5 Scaled coefficient of six drivers of biodiversity change in each region within an autoregressive error model (NSR: native species richness; RBA: river basin area; DEM: the mean elevation of each geographic unit; ORD: Strahler stream-order; LEN: weighting the river order within each geographic unit by reach length; DOF: degree of fragmentation; USE: consumptive water use; URB: urbanization index). Number of units used in the models: Lake (n = 23); Tributary (n = 17); Mainstream (n = 16). ***$P < 0.001$, **$P < 0.01$, *$P < 0.05$

fish biodiversity in the middle reaches of the Yangtze River is mainly the extensive shoreline modification, and hydraulic engineering construction that have disrupted the connectivity of the river (Xiong et al. 2021). The least impacted units (CCBF = 6) occurred in high-latitude areas and several units within the lake region, where weaker anthropogenic pressures and stronger local management (e.g., seasonal bans, wetland reserves) helped buffer biodiversity losses (Teng et al. 2020). The relatively smaller change in fish diversity in the Lake region, compared to the Mainstream and Tributary regions, stems from both the ecological and institutional factors. Natural hydrological connectivity, greater habitat heterogeneity, and semi-enclosed system dynamics provide refuges that maintain local populations within the Poyang and Dongting Lake regions. In these areas, pre-existing conservation actions, including wetland nature reserves, acted as partial safeguards, whereas the open river network remained exposed to continuous shipping, noise, and hydrological disturbance (Teng et al. 2020). This heterogeneity of human influence explains the spatial gradient in CCBF values (Fig. 4B).

During the last decades, fish assemblages in the Yangtze River Basin have undergone a pronounced decline in local taxonomic, functional, and phylogenetic α-diversity (TR, FR, and PR). Although only a few species were pushed to extinction over the whole river basin during that period, see (Zhang et al. 2020a) for *Psephurus gladius*, we here report biodiversity declines in most of the 56 considered geographical units, with 77–84% of the units experiencing a decline in TR, FR or PR. Among the most impacted species are the long-migrating and slow life-history species, such as the critically endangered Chinese sturgeon (*Acipenser sinensis*) (Liu et al. 2019). The primary reason for the TR decrease is the high rate of species disappearance in most units, surpassing the introduced species (Table S4). Moreover, the species that were extirpated from units had unique functional traits, leading to a decrease in FR. The decline in PR is probably attributed to the disappearance of distantly related species and the introduction of closely related species (Winter et al. 2013). An exception emerged in several high-altitude units (2–4) of the upper Jinsha River, where high TR, FR, and PR can be linked to human-mediated species introductions rather than ecological recovery.

In contrast to the basin-wide decline in local α-diversity, β-diversity (TD, FD, and PD) increased in more than 85% of units (Fig. 2D–F), revealing increasing regional differentiation rather than homogenization. Given that many units experienced concurrent species losses, this increase in β-diversity primarily reflects uneven local disappearances of distinct taxa. This within-basin heterogenization contrasts with the global trend of inter-basin homogenization (Su et al. 2021; Toussaint et al. 2014; Villéger et al. 2011), suggesting that anthropogenic disturbance

does not necessarily lead to biotic homogenization and that its direction is strongly scale-dependent, underscoring the need for finer-resolution assessments. Mechanistically, the concurrent α-diversity loss and β-diversity gain could be interpreted as an outcome of spatially heterogeneous environmental filtering and selective extinction (Rolls et al. 2023). Fragmentation, altered sediment loads, and declining water quality filtered out habitat- or flow-sensitive taxa in each sub-basin. Meanwhile, different generalist or stress-tolerant species—each adapted to local combinations of disturbances—expanded in different units, leading to increased spatial differentiation in functional composition (e.g., higher functional dissimilarity) rather than homogenization. Overall, the observed divergence likely reflects biotic reassembly under contrasting anthropogenic constraints rather than ecological recovery. Consistent with this pattern, the Three Gorges Dam located in the middle Mainstream region near Yichang likely contributed to the high CCBF values observed there. Upstream units (above Yichang) experienced reservoir-induced shifts from lotic to lentic conditions, favouring generalist and limnophilic species, whereas downstream units faced altered flow pulses and sediment starvation (Liu et al. 2019). These divergent pressures may explain the high spatial heterogeneity in β-diversity change within the middle reach.

Environmental drivers of CCBF

The CCBF scores showed a positive correlation with human activity linked to industrialization and economic development, including the degree of fragmentation (DOF), consumptive water use (USE), and urbanization (URB) (Fig. 5). This finding is attributed to the development of the Yangtze River Economic Belt and the population growth in recent decades, which led to an increase in the degree of fragmentation, consumptive water use, and urbanization (Grill et al. 2019; Xiong et al. 2023). Habitat fragmentation is one of the main negative outcomes of long-term dam construction activities, hindering aquatic animal movements, altering natural flow patterns, and disrupting habitats and nutrient supply (Trovillion et al. 2023). Limited or no connectivity between rivers and floodplains inhibits the ability of aquatic organisms to spawn, feed, and find shelter, leading to an overall decline in biodiversity (Liermann et al. 2012; Pettit et al. 2017). Thus, habitat fragmentation caused by river channel development and dam construction is one of the main drivers of CCBF increase in the Mainstream and Tributary regions. In the Lake region, the extension of seasonal dry periods and the increased water use for agricultural and industrial purposes have made consumptive water use an important anthropogenic driver of CCBF (Fig. 5). Moreover, road paving and infrastructure construction in urban areas lead to artificial canalization and embankment construction of rivers, ultimately resulting in river degradation and loss of ecosystem functions (Kern and Langerhans 2018; Marchetti et al. 2006). Accordingly, URB emerged as an important factor influencing CCBF in tributary regions of the Yangtze River Basin. These results suggest that the various anthropogenic factors influencing fish biodiversity in the Mainstream, Tributary, and Lake regions account for the inconsistent changes in CCBF across the Yangtze River Basin.

Meanwhile, the relationship between CCBF and native species richness (NSR) exhibited pronounced spatial heterogeneity. In both the Lake and Tributary regions, native species richness was positively correlated with CCBF, suggesting that units with higher native richness tended to experience larger biodiversity changes. A plausible explanation is that, in upstream and mid-altitude tributaries where historical native diversity is high, many endemic mountain fishes occupy narrow thermal and/or flow niches, making them particularly sensitive to even moderate disturbances. In the lower mainstream and lake systems, intense human activities (e.g., urbanization, land reclamation, and water extraction) have substantially altered habitat complexity and water quality, potentially driving severe biodiversity loss in areas that originally supported rich assemblages. In contrast, the negative correlation between native species richness and CCBF in the Mainstream region indicates that fish assemblages in high-diversity mainstream units experienced smaller diversity changes than those in low-diversity units. This pattern may reflect either greater disturbance resistance in species-rich assemblages and/or lower exposure to anthropogenic pressures in these units. For instance, some upstream reaches—particularly high-elevation sections—tend to experience lower human pressure and greater conservation coverage than the middle and lower

reaches, where urbanization and industrial–agricultural activities are more intensive (Zhu et al. 2025). Notably, a significant negative correlation between length-weighted river order (LEN) and CCBF was observed only in the Tributary region, but not in the Mainstem or Lake regions. This suggests that in tributaries, higher-order (e.g., larger and more connected) river segments are associated with greater biodiversity change, likely because they act as cumulative corridors for multiple stressors (Muneepeerakul et al. 2008). In contrast, the absence of this relationship in other regions implies that the overwhelming impacts of large-scale, localized disturbances (e.g., dams in the mainstem, eutrophication in lakes) have masked the subtler role of geomorphic structure.

Our study has several limitations. First, we acknowledge that commercial overfishing has historically been a major driver of fish depletion in the Yangtze River, as also recognised by previous studies (Zhang et al. 2020b). However, fishing intensity could not be retained in the final model due to high multicollinearity with other anthropogenic variables ($r > 0.85$, Fig.S2), reflecting the historically simultaneous intensification of multiple human activities in the Yangtze River Basin. Second, several potentially important drivers—including sand mining, intensive shipping, and climate change—were not directly quantified owing to data limitations at the basin-wide scale over the 1976–2020 period. Third, native species richness was included as a covariate following the CCBF framework, but we acknowledge that it could also be viewed as a response variable. Future studies with higher-resolution data and longer time series are needed to disentangle the relative contributions of these missing drivers.

Implications for conservation management

Understanding spatial patterns of fish biodiversity across multiple dimensions is essential for guiding effective management and conservation strategies for the Yangtze River Basin. Traditional univariate or taxonomic diversity indices alone often fail to capture the full spectrum of anthropogenic impacts because ecological responses differ across functional and phylogenetic facets and spatial scales (Sinclair et al. 2024). Our multidimensional assessment, integrating taxonomic, functional, and phylogenetic information through the Cumulative Change in Biodiversity Facets (CCBF) index, provides a more comprehensive evaluation of ecosystem alteration. The high CCBF values (all exceeding 6) throughout the Yangtze River Basin indicate the severe and widespread impact of human activities. These findings advocate for the use of integrated indices like CCBF in prioritizing conservation actions. Specifically, we propose a stratified management framework tailored to the dominant pressures in each region. In the middle mainstream sub-basins—identified as priority areas due to their high CCBF scores—conservation efforts should focus on mitigating habitat fragmentation by improving hydrological connectivity through fish passage construction and strategic environmental flow management. Simultaneously, in the Lake region, where consumptive water use is a key driver, policies must enforce sustainable water withdrawal quotas, especially during seasonal dry periods, and promote water-efficient irrigation practices. For Tributary systems under intense urbanization, measures should emphasize riparian habitat restoration, the reduction of river channelization, and the implementation of constructed-wetland systems to treat urban runoff and mitigate nutrient and contaminant inflows. At the basin scale, while the ongoing 10-year fishing ban (2021–2030) provides a critical regulatory foundation, our findings emphasize that species protection alone is insufficient without concurrent habitat and hydrological restoration. Hence, integrating flow management, habitat connectivity, and water-quality governance within a coordinated, region-differentiated conservation framework will maximize ecological recovery (Feng et al. 2023).

Although our analysis ends in 2020, the subsequent full implementation of the Ten-Year Fishing Ban (starting January 2021) and ongoing habitat restoration efforts have already shown initial signs of fish recovery in the Yangtze River. A recent study reported that the ban has successfully halted a 70-year decline in fish resources, with marked improvements in fish biomass, body condition, species diversity, and initial signs of recovery among threatened species (Xiong et al. 2026). In the middle reaches of the Ynagtze River, the fishing ban has improved fish body size and population structure of species, though species diversity recovery remains uneven across different sites, indicating that full recovery may require extended conservation efforts (Du et al.

2026). Beyond the fishing ban itself, targeted habitat restoration projects have played a complementary role. One notable example is the connectivity restoration project for the Chishui River tributary, which removed or modified in-stream barriers to facilitate fish migration, resulting in a detectable increase in fish species richness and the emergence of previously rare endemic species (Liu and Liu 2023). These combined measures—fishing ban, habitat restoration, and species reintroduction—have synergistically contributed to the initial recovery of fish biodiversity. Nevertheless, recovery remains spatially heterogeneous, and full restoration of the Yangtze River's freshwater fish biodiversity will require sustained, long-term conservation efforts (Ministry of Agriculture and Rural Affairs of China 2025). Our study provides a critical pre-ban multidimensional baseline against which these post-ban recovery trends can be evaluated. We strongly encourage continued application of the CCBF framework to track post-ban biodiversity trajectories across taxonomic, functional, and phylogenetic dimensions.

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Author contribution J.X., G.S., K.Z. and J.D. conceived the idea. G.S., J.D., and K.Z. contributed to the study design. J.D., G.S., K.Z. and J.X. contributed to the acquisition and analysis of data. J.D., K.Z., S.B. and G.S. contributed to the interpretation of results. J.D., K.Z., G.S., S.B., D.C., M.L., L.Y. and X.D. wrote and edited the manuscript.

Data availability The study's raw data used is freely accessible to the public ("Methods" section). Supplementary Information contains the processed data that was produced for this study.

Declarations

Competing interests The authors declare no competing interests.

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